

DEPARTMENT OF MECHANICAL ENGINEERING
MBM UNIVERSITY, JODHPUR
MID TERM-I: - 4ME42A(ME): METAL CUTTING & METROLOGY

Dt: FEB. 03, 2025

TIME: 2:30-3:30PM

MM: 20

Q1 (CO1). Explain theory of chip formation and their types. (5)

Q2 (CO1). Derive the expression for Shear angle and find the relationship among velocities in metal cutting. (5)

Q3 (CO2). During turning a carbon steel rod of 160 mm diameter by a carbide tool of geometry 0, 0, 10, 8, 15, 75, 0 (mm) at speed of 400 rpm, feed of 0.32 mm/rev. and 4.0 mm depth of cut, the following observations were made:

Tangential component of the cutting force, $f = 100$ N

Axial component of the cutting force $F = 800$ N

Chip thickness, $t = 0.8$ mm

For the above machining condition determine the values of

- a) Friction force F , normal force N acting at the chip-tool interface
- b) Yield strength of the work material under this machining condition
- c) Cutting power consumption in kW.
- d) Draw Merchant Circle Diagram for the above machining process.

(10)

MBM University, Jodhpur
Department of Mechanical Engineering
B.E. IInd Year IIIrd Semester
First Mid-Term Examination Session 2024-25
3ME43A(ME): Kinematics of Machines

Time: 1 hour

Maximum Marks – 10

Q. 1(a). In a slider-crank mechanism, the lengths of the crank and connecting rod are 200 mm and 800 mm respectively; crank has turned 30° . Determine the velocity of slider and the angular acceleration of connecting rod if crank rotates at 40 rad/sec (counter clockwise direction).

(4)

(b) State Arnold-Kennedy theorem of three centres.

(1)

Q. 2 Define kinematic link, kinematic pair and kinematic chain.

(1)

(b) Sketch and explain the various inversion of double-slider crank chain mechanism.

(3)

(c) Explain Gruebler's criterion for degree of freedom for planar mechanisms.

(1)

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BE II Yr. III SEM.

Material Technology

Paper Code- 3ME42A(ME)

Mid Term Exam.-I

Date: 25/09/24

Max. Marks: 10

All questions compulsory

Q.1	CO1	Describe different crystal structure with diagram.	3
Q.2	CO1	Describe different crystal imperfection with diagram.	3
Q.3	CO1	Copper has F.C.C. structure and the atomic radius $1.278 \text{ \AA}$. Calculate the density of copper crystal. Given atomic weight of copper = 63.5. Avogadro no. = 6.023×10^{23} .	4

First Mid Term Exam 2024
Class BE II Year III Sem.
Mechanical Engineering Department
Subject: 3ME21A(MA) : Mathematics

Time: 1 hour

MM: 1

Note Do any two questions. All questions carry equal marks.

1. (a) solve. $y = a^2 t$ by Monge's Method
(b) solve $tdx = (t-2x) dt$; $tdy = (tx + ty + 2x - t) dt$
2. (a) solve $(y^2 + yz) dx + (xz + z^2) dy + (y^2 - xy) dz = 0$
(b) solve by charpit's method $(p^2 + q^2) y = qz$
3. (a) Use Newton Gregory forward difference interpolation formula to compute $y(3.62)$ given $(x, y) = (3.60, 36.598), (3.65, 38.475), (3.70, 40.447), (3.75, 42.521)$
(b) Use Lagranges interpolation formula find $y(9.5)$ given $(x, y) = (7, 3), (8, 1), (9, 1), (10, 9)$.